# DERIVED LIMITS OF INVERSE SYSTEMS WITH SURJECTIVE TRANSITION MAPS ON COHOMOLOGY

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Higher cohomology of a quasi-coherent module on an affine open vanishes. invoked at proof lines 335-337. Verifies the vanishing condition of the suitable-system lemma.
- On an affine open the sections functor identifies quasi-coherent modules with modules, so a surjective map of modules gives a surjective map on sections. invoked at proof lines 338-340. Verifies the surjectivity condition of the suitable-system lemma.
- The derived limit lemma for suitable inverse systems computes H^q(Rlim K_n) as lim H^q(K_n). invoked at proof lines 341-343. Proves part (1).
- The comparison lemma between inverse limits and derived limits of module systems gives parts (2) and (3). invoked at proof lines 344-345. Proves parts (2) and (3).

*Source reference.* Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cited at proof lines 336-337 to verify the vanishing condition on affine opens; the conclusion is inside the counted span.

*Source reference.* Original lemma 01IB: sources/stacks-chow/tags.txt, source lines 1972–1972. Cited at proof lines 339-340 for the surjectivity of the maps on sections; the conclusion is inside the counted span.

*Source reference.* Original lemma 0BKU: sources/stacks-chow/tags.txt, source lines 14572–14572. Cited at proof lines 341-343 for part (1) after its conditions are verified inside the counted span.

*Source reference.* Original lemma 0BKS: sources/stacks-chow/tags.txt, source lines 14570–14570. Cited at proof lines 344-345 for parts (2) and (3); the conclusion is inside the counted span.

**Lemma 0.2.** *Let $X$ be a scheme. Let $(K_n)$ be an inverse system of $D_{QCoh}(\mathcal{O}_X)$ with derived limit $K = R\lim K_n$ in $D(\mathcal{O}_X)$. Assume $H^q(K_{n+1}) \to H^q(K_n)$ is surjective for all $q \in \mathbf{Z}$ and $n \geq 1$. Then*

(1) $H^q(K) = \lim H^q(K_n)$,

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(2) $R\lim H^q(K_n) = \lim H^q(K_n)$, *and*
(3) *for every affine open* $U \subset X$ *we have* $H^p(U, \lim H^q(K_n)) = 0$ *for* $p > 0$.

*Proof.* Let $\mathcal{B}$ be the set of affine opens of $X$. Since $H^q(K_n)$ is quasi-coherent we have $H^p(U, H^q(K_n)) = 0$ for $U \in \mathcal{B}$ by Cohomology of Schemes, Lemma 01XB. Moreover, the maps $H^0(U, H^q(K_{n+1})) \to H^0(U, H^q(K_n))$ are surjective for $U \in \mathcal{B}$ by Schemes, Lemma 01IB. Part (1) follows from Cohomology, Lemma 0BKU whose conditions we have just verified. Parts (2) and (3) follow from Cohomology, Lemma 0BKS. □